Browser Manager - Google Admin Deployment Guide

This guide walks you through deploying the **Browser Manager** extension to all users in your organization via Google Workspace Admin Console.

Prerequisites

✅ **Google Workspace admin access** with permissions to manage Chrome apps & extensions
✅ **Chrome Web Store Developer account** ($5 one-time registration fee)
✅ **Extension ready to deploy** (this repo contains the complete, tested extension)

Deployment Options

You have **two deployment paths**:

Option A: Chrome Web Store (Recommended)

Upload to Chrome Web Store as **unlisted** (only users with the link can find it) or **private** (only your domain can install it). Google Admin can then force-install it by extension ID.

Pros: Easiest to manage, automatic updates, Google-hosted
Cons: Requires Chrome Web Store developer account, $5 fee, review process (usually 1-3 days)

Option B: Self-Hosted (Advanced)

Host the .crx file on your own server and configure Admin Console to install from your URL.

Pros: No Chrome Web Store account needed, full control
Cons: More complex, you manage updates manually, requires HTTPS hosting

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Option A: Chrome Web Store Deployment (Step-by-Step)

Step 1: Package the Extension

1. **Set your admin password** (if you haven't already):

- Edit popup.js line 32: const ADMIN_PASSWORD = "YOUR_PASSWORD_HERE";
- Edit options.js line 13: const ADMIN_PASSWORD = "YOUR_PASSWORD_HERE";

2. **Create a .zip file** of the extension:

```
# From your project directory
zip -r mck-browser-manager.zip * -x "*.git*" "*.DS_Store" "node_modules/*"
```

Or manually:

- Select all files in /mck-browser-manager/ folder
- Right-click → **Compress** (Mac) or **Send to → Compressed folder** (Windows)
- Name it mck-browser-manager.zip

⚠️ **Important:** Zip the **contents** of the folder, not the folder itself. When you unzip, you should see manifest.json at the root, not mck-browser-manager/manifest.json.

Step 2: Chrome Web Store Developer Registration

1. Go to Chrome Web Store Developer Dashboard
2. Sign in with your Google Workspace admin account (or a dedicated developer account)
3. **Pay the $5 one-time registration fee** (if not already registered)
4. Agree to the Developer Agreement

Step 3: Upload Your Extension

1. Click **"New Item"** in the Developer Dashboard
2. **Upload** your mck-browser-manager.zip file
3. Fill in the required listing information:

Store Listing Tab:

- **Name:** Browser Manager
- **Summary:** Enterprise browser management for your organization — one-click cache clearing, performance tuning, and privacy controls.
- **Description:**

  Browser Manager is an enterprise Chrome extension for employees, providing centralized browser management tools.

  Features:
  • One-click cache clearing with visual feedback
  • Auto-clear cache on Chrome startup (IT-managed)

• Performance controls: Memory Saver, Tab Hibernation, Preload Pages
• Privacy settings: Do Not Track, 3rd Party Cookie blocking, WebRTC protection
• Centrally managed by IT Help Desk

This extension is designed for internal use and managed via Google Admin Console.

- **Category:** Productivity
- **Language:** English (United States)

Upload screenshots (recommended):

- Take 2-3 screenshots of the extension popup (Cache tab, Privacy tab, etc.)
- Chrome Web Store requires at least 1 screenshot (1280x800 or 640x400)

Privacy Tab:

- **Single Purpose:** Browser management and performance optimization for enterprise users
- **Permission Justification:** Explain each permission:
  - browsingData: Required to clear cache
  - storage: Persist user settings
  - tabs: Memory Saver and Tab Hibernation features
  - privacy: Apply Do Not Track and WebRTC settings
  - contentSettings: Block 3rd party cookies
  - alarms: Schedule automated tab management

Visibility:

- Select **"Unlisted"** (users with the link can install, but it won't appear in search)
- OR **"Private"** (only your Google Workspace domain can install) — requires domain verification

4. Click **"Submit for Review"**

Step 4: Wait for Approval

- **Review time:** Usually 1-3 business days (sometimes as fast as a few hours)
- You'll receive an email when the extension is approved
- Once approved, note the **Extension ID** (looks like: abcdefghijklmnopqrstuvwxyz123456)

Step 5: Deploy via Google Admin Console

Once your extension is live in the Chrome Web Store:

1. **Sign in to Google Admin Console**
   Go to admin.google.com
2. **Navigate to Chrome Management**
   Devices → Chrome → Apps & extensions
3. **Add the Extension**
   - Click **"Add Chrome app or extension"** (yellow ➕ button)
   - Paste your **Extension ID** from the Chrome Web Store
   - Click **"Select"**
4. **Configure Installation Policy**
   - **Who:** Select the organizational unit (OU) to deploy to:
     - Select **"All users"** for company-wide deployment
     - Or choose specific departments/groups
   - **Installation Policy:** Choose **"Force install"**
     - This automatically installs the extension for all users in the selected OU
     - Users **cannot** uninstall it (true enterprise policy enforcement)
   - **Installation URL:** Auto-filled (Chrome Web Store URL)
5. **Pin to Toolbar (Optional but Recommended)**
   - Under "User Experience" settings:
     - Set **"Toolbar pin"** to **"Force pinned"**
     - This ensures the extension icon is always visible in the Chrome toolbar
6. **Configure Settings (Optional)**
   You can also pre-configure settings via Admin Console policies, but since your extension uses `chrome.storage.local`, users will have the default settings defined in `background.js`.
7. **Click "Save"**

Step 6: Rollout

- **Propagation time:** Changes can take up to 24 hours to apply to all devices
- **For faster testing:** Have a user:
  1. Close and reopen Chrome
  2. Or visit `chrome://policy` → Click **"Reload policies"**

- The extension will automatically install on all managed Chrome browsers within your selected OU
- Users will see “Browser Manager” in their toolbar (if pinned) or extensions list

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Option B: Self-Hosted Deployment (Advanced)

If you don’t want to publish to Chrome Web Store:

Step 1: Package and Sign the Extension

1. **Create a .crx package** (Chrome Extension Package):

   ```
   # Chrome requires extensions to be signed
   # Use chrome://extensions/ → Pack extension
   # Or use command line (requires Chrome binary)
   ```

2. **Host the .crx and update.xml files** on an HTTPS server:

   - Upload mck-browser-manager.crx to your web server
   - Create an update.xml manifest (example below)

   update.xml:

   ```
   <?xml version='1.0' encoding='UTF-8'?>
   <gupdate xmlns='http://www.google.com/update2/response'
   protocol='2.0'>
     <app appid='YOUR_EXTENSION_ID'>
       <updatecheck codebase='https://your-domain.com/mck-browser-
   manager.crx' version='1.0.0' />
     </app>
   </gupdate>
   ```

Step 2: Deploy via Admin Console

1. In Google Admin Console:
   Devices → Chrome → Apps & extensions → Add Chrome app or extension
2. **Choose “Add custom app”**
3. Enter:
   - **Extension ID:** (from your .pem file or generated during packing)
   - **Update URL:** https://your-domain.com/update.xml
4. Set **Installation Policy** to **“Force install”**
5. Click **“Save”**

⚠️ **Self-hosting drawbacks:** - More complex initial setup - You must manually update update.xml and .crx for every new version - No automatic security/malware scanning by Google

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Updating the Extension

If using Chrome Web Store:

1. Make your code changes locally
2. Increment the version in manifest.json (e.g., 1.0.0 → 1.0.1)
3. Re-zip the extension
4. Upload to Chrome Web Store Developer Dashboard (same listing, click "Package" tab → "Upload new package")
5. Submit for review
6. Once approved, updates **automatically roll out** to all installed instances (within 24-48 hours)

If self-hosting:

1. Update your code
2. Increment version in manifest.json
3. Re-create the .crx file
4. Update update.xml with the new version number and .crx URL
5. Upload both files to your server
6. Chrome checks for updates periodically (every few hours)

Testing the Deployment

Before full rollout:

1. **Create a test OU** in Google Admin:
   - Admin Console → Directory → Organizational units → **Create test OU**
   - Add 1-2 test users to this OU
   - Deploy the extension to this OU first with "Force install"
2. **Test user workflow:**
   - Sign in to Chrome as a test user
   - Verify the extension auto-installs
   - Check all features:
     - Cache clearing works
     - Admin password gate works (try toggling "Auto-clear on startup")
     - Advanced Settings cog reveals tabs
     - Privacy/Performance toggles apply correctly

3. **Once validated:**
   - Expand deployment to **All users** or your main OUs

Troubleshooting

Extension not installing for users?

- Check the OU selection in Admin Console (ensure the user is in the target OU)
- Verify "Force install" is selected (not "Allow install")
- Ask users to visit chrome://policy → **Reload policies** → check "ExtensionInstallForcelist"
- Check Chrome Management enrollment: chrome://management/ should show "Your browser is managed by [your-domain]"

Users can uninstall the extension?

- Ensure you selected **"Force install"** (not "Allow install") in Admin Console
- Force install prevents uninstallation by users

Settings not persisting?

- The extension uses chrome.storage.local which is tied to the Chrome profile
- If users are using temporary/guest profiles, settings won't persist
- Ensure users are signed in to their Workspace account in Chrome

Admin password visible in source code?

- **This is intentional and expected for client-side extensions** — the password is a deterrent, not a security boundary
- For true enforcement, use Chrome Enterprise Policies (can lock settings via Admin Console)
- The current password gate is sufficient for preventing **accidental** changes by non-technical users

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Security Best Practices

1. **Restrict Admin Console access** to IT staff only
2. **Review extension updates** before deploying (if self-hosting)
3. **Monitor user feedback** via IT Help Desk link (support.example.com)
4. **Use "Force install"** to prevent users from disabling/removing the extension
5. **Change the admin password** periodically (update in both popup.js and options.js, re-deploy)

Support & Maintenance

- **IT Help Desk:** https://support.example.com
- **Extension version:** Displayed in footer of popup and options page
- **Chrome version compatibility:** Manifest V3 requires Chrome 88+ (2021), all modern Chrome browsers supported

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Quick Reference

Task	Google Admin Console Path
Deploy extension	Devices → Chrome → Apps & extensions → Add app
Pin to toolbar	Extension settings → Toolbar pin → Force pinned
View installed extensions	Devices → Chrome → Apps & extensions
Check user devices	Devices → Chrome → Devices
Force policy update	User visits chrome://policy → Reload policies

Additional Resources

- Chrome Enterprise Policy List
- Chrome Web Store Developer Documentation
- Google Workspace Admin Help: Manage Chrome Apps

Questions? Contact your IT Help Desk or the extension developer.